

# Design and Deployment of an Imaging Spectrometer for the Characterisation of Spatial Chirp



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# Chapter 1

## Introduction

### 1.1 Plasma Modulated Plasma Accelerator

A plasma can be formed by ionising a gas target using a high-intensity, short (10 fs) laser pulse. The interaction of the laser pulse with the plasma can form accelerating structures with field strengths of the order of 100 GV/m. In a laser-wakefield accelerator scheme (LWFA), these accelerating structures may be used to achieve electron energies in the order of 100s MeV to GeV [1]. Such schemes may lead to the production of compact, low cost electron sources with applications such as light source generation [2], QED experiments [3], radiobiology studies [4], and probing fundamental plasma physics [5].

LWFA schemes are currently limited by the repetition rates of their laser systems, most commonly  $\lesssim 10$  Hz with Ti:sapphire lasers. The University of Oxford's Laser Plasma Accelerator Group (LPAG) is working with collaborators on the plasma-modulated plasma accelerator (P-MoPA) project to address this [6, 7, 8, 9, 10]. The project aims to use thin-disk lasers with high repetition rates ( $\sim 1$  kHz) to drive a LWFA electron source. The pulse duration of such lasers is too long to drive a LWFA, so the following scheme is used to create a pulse-train (see Figure 1.1): A short, low-energy seed-pulse is co-propagated with a long drive-pulse through a modulator stage containing a gas. The seed-pulse ionises the gas and drives a low-intensity plasma wave that modulates the drive-pulse, producing side-bands in its spectrum. A compressor then removes the phase on these side-bands, converting the picosecond pulse into a train of short pulses spaced temporally by the plasma period. This pulse train can then be directed into a second plasma stage of the same density, resonantly exciting a wakefield into which electrons can be injected and subsequently accelerated. This modulation-compression process is sensitive to the drive-pulse's spatio-spectral content; any spatio-spectral couplings would vary the modulation across the pulse's

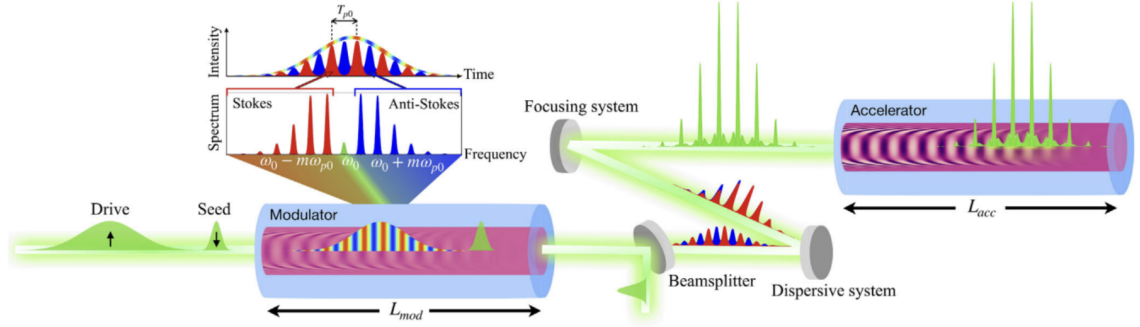


Figure 1.1: The Plasma Modulated Plasma Accelerator (P-MoPA) scheme. Reproduced from [6].

profile and degrade the resulting pulse-train, weakening the pulse-train’s ability to drive a wakefield and leading to unstable electron acceleration. The characterisation of spatio-spectral couplings in the thin-disk lasers used in the P-MoPA project is thus essential for the project’s success.

## 1.2 Angular and Spatial Chirp

### 1.2.1 Chirped Pulse Amplification

Prior to Gérard Mourou and Donna Strickland’s invention of chirped pulse amplification (CPA) [11], for which they were awarded the Nobel Prize in Physics in 2018, it was extremely challenging to produce ultrashort laser pulses with enough intensity for modern LWFA without damaging the optical equipment and introducing non-linear effects. CPA avoids these issues through a staged amplification procedure (see Figure 1.2): An ultrashort, low-amplitude pulse enters a pulse stretcher, composed of two identical parallel diffraction gratings. This stretches the pulse temporally, reducing its intensity without loss of energy. The pulse is subsequently amplified then directed through a pulse compressor, similar to the pulse stretcher, which compresses the pulse back to its original duration. The result is a high-intensity ultrashort pulse, but crucially this intensity is only reached post-compressor, leaving all previous optical components unharmed.

As with most real optical systems, CPA systems are highly sensitive to misalignments. In particular, a relative misalignment between the two gratings of the stretcher or the compressor, or a mismatch between the dispersive properties of the stretcher and the compressor, may produce spatio-spectral couplings in the pulse. One such coupling is known as *angular chirp*, a correlation between the angle of propagation

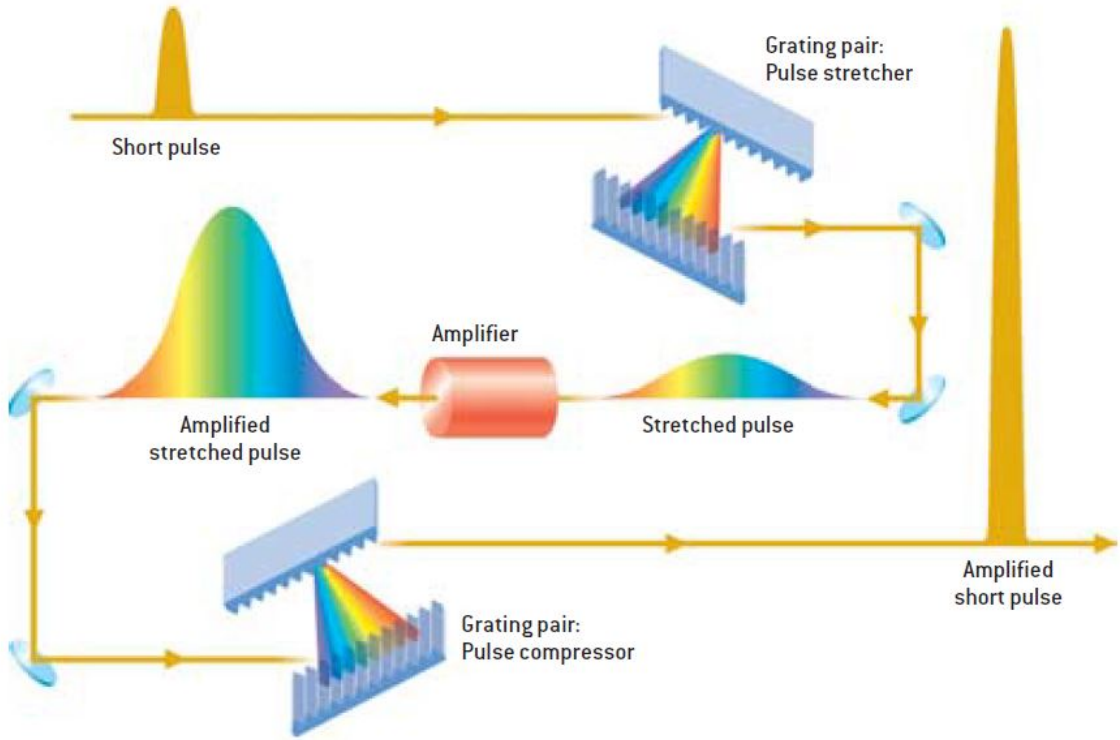


Figure 1.2: Illustration of Chirped Pulse Amplification (CPA). Reproduced from [12].

$\theta$  and wavelength  $\lambda$  within a pulse of finite bandwidth. When an angularly chirped pulse propagates over a finite distance, it acquires *spatial chirp*, a correlation between position  $x$  along an arbitrary transverse axis across a pulse and wavelength  $\lambda$ .

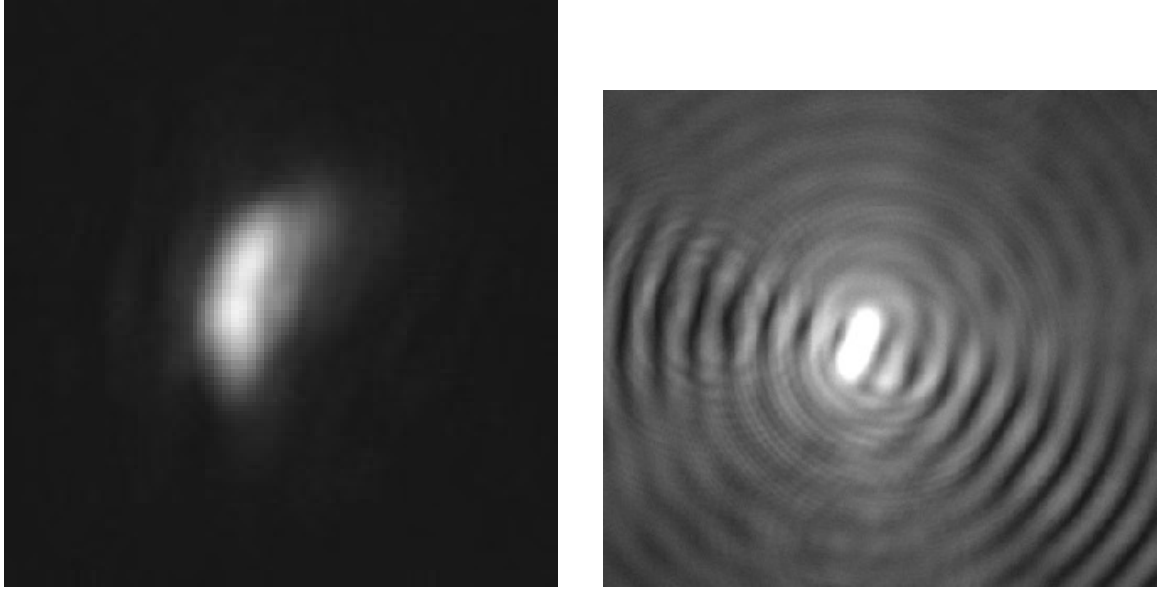
### 1.2.2 Angular-to-Spatial Chirp Conversion at Focus

A useful property of the converging lens is the conversion between angular and spatial information at its focal plane. In particular, any collimated plane wave incident on an ideal thin lens of focal length  $f$  is, in the paraxial approximation, focussed to a transverse position  $x$  on the lens' back focal plane proportional to its angle of incidence  $\theta$ :

$$x = f\theta. \quad (1.1)$$

Consequently, if the incident pulse is angularly chirped, meaning  $\theta$  is spectrally dependent, the transverse position  $x$  of the resulting focal spot also becomes spectrally dependent, meaning it is spatially chirped. The result is a focal spot that is stretched along an arbitrary transverse axis, as opposed to a typical circular focal spot. (see Figure 1.3).





(a) Lens focus

(b) Axicon focus

Figure 1.3: Spatially chirped foci resulting from angularly chirped incident pulses. Images courtesy of Alexander Harrison.

Differentiating Equation 1.1 with respect to wavelength  $\lambda$ :

$$\frac{dx}{d\lambda} = f \frac{d\theta}{d\lambda}. \quad (1.2)$$

The derivative  $\zeta \equiv \frac{dx}{d\lambda}$  is known as *spatial dispersion*, corresponding to a pulse's spectrally dependent transverse position  $x(\lambda)$ , and is the conventional metric used for describing spatial chirp throughout this project. Another common metric used for describing spatial chirp is known as the frequency gradient,  $\frac{d\omega}{dx}$ , corresponding to a pulse's transverse-positionally dependent angular frequency  $\omega(x)$ . While these parameterisations both describe spatial chirp, they are not simply reciprocals of each other [13]. Spatial dispersion is used throughout this project because of its convenient relationship with the angular dispersion  $\frac{d\theta}{d\lambda}$  that produces it. Wavelength is chosen as the spectral unit throughout this project due to its direct relationship with the diffraction angle of optical gratings.

1.2.3 Consequences for the Plasma Wave

1.3 Measuring Angular Chirp

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1.5 Report Outline

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## Results

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# Chapter 8

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# Chapter 9

## Conclusions and Future Work

### 9.1 Conclusions

### 9.2 Future Work

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